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Venue entry payment system for vehicles

Abstract

There is a system for admitting automobiles to a venue comprising: at least one camera for recognizing an automobile, and a license plate. A server is configured for receiving a picture of the automobile and a picture of the license plate of the automobile. Furthermore, there is at least one database for storing, matching and identifying the picture of the automobile and the license plate. In addition, there is at least one screen for allowing a user to identify their automobile taken from a list of automobiles. There is also a process for tracking an automobile within an area which comprises the steps of identifying a pre-defined area and identifying an automobile for screening into the area. Next the auto and its license plate are photographed. This information is stored in a database and then used to generate invoices and track the location of these automobiles. Customer can access the database, enter the license plate of his vehicle, the database recognizes his vehicle, the customer makes payment for his vehicle, and the app places the customer information onto the PAID category list on the database.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a non-provisional application that claims priority from U.S. Provisional Application Ser. No. 63/092,725 filed on Oct. 16, 2020, the disclosure of which is hereby incorporated by reference.

BACKGROUND OF THE INVENTION

(1) At least one embodiment of this invention relates to a system for payment of entry fee for motor vehicles into fee areas such as parks, campgrounds and beaches without the requirement for gated

entry, payment booths or manual labor at point of entry.
(2) Current entry systems require the use of entry gates and entry booth attendants at all entrances for the collection of entry fees. The variable and sometimes very high entry demand results in long lines of cars waiting to get to the toll booth to gain entry. Additionally, the need to carry out a face-to-face transaction at the gate can pose health risk during epidemics such as covid19 and influenza. Therefore, a more efficient and non-contact entry transaction is desirable.
(3) During peak periods, it is common to have over 100 cars in line waiting to get to the toll booth.
(4) For example, in a news report by Channel 4 NBC NY, on Apr. 20, 2020, a reporter stated that "Around 30,000 people were estimated to have gone to Robert Moses State Park beach on Long Island on Saturday". These waiting lines sometimes extend onto the feeder highways and pose traffic problems as well. At an average processing speed of 30-60 seconds per car, the 100th car could see a waiting time of over an hour for entry. It's also not possible apriori to know the length of this waiting time until a customer arrives at the location and observes the long lines.
(5) Staff at these locations, at the hourly wage of $15-20 and whose only function is to collect the entry fee, can account for 10 percent or more of the venue operating cost. It is therefore desirable to have an entry system which minimizes or eliminates congestion and the need for manual labor at point of entry.
(6) Commercial license plate reading systems (LPC) are available which read car license plates at points of entry. Experience and the technical literature show that these systems operate at about 75 to 85 percent recognition rate leaving up to a quarter of entering vehicles unread and unidentified. See for example a report by D J Roberts and M Casanova titled "Automated License Plate Recognition Systems: Policy and Operational Guidance for Law Enforcement" published in September 2012 and available as document #239604 from the National Criminal Justice Reference Service. Recent improvements in LPC technology have increased the recognition rate to about 95 percent by incorporating machine learning boards into these systems. An article by Tom Simonette on Jan. 27, 2020 in WIRED magazine reports that "police in Sands Point, New York, a gated village on the north shore of long Island, that police chief Thomas Ruehle says new Rekor software on his departments LPC system correctly reads plates at about 97 percent of the time compared to old technology which operated at around 80 percent." These improved systems still leave about 3 to 5 percent of incoming cars unidentified. Sands Point is a gated community with limited access routes and therefore is an unusually controlled and favorable environment for LPC system operation. Most experiences with these smartboards coupled systems show recognition efficiencies in the low to mid-ninety percent range and are therefore not reliable as stand-alone cashiering systems. Substantial performance improvements are required for these, otherwise attractive, systems to become useful and reliable in these applications.
(7) LPC reading problems come from a wide variety of factors such as lighting, sun position, contrast issues, dirty, defective or partially obscured plates, changing weather conditions and other issues. Therefore, many issues will need to be addressed in order to further improve reading rates meaningfully. It is likely that further progress will be incremental and modest. These reading errors can further add to the waiting time at the entry point and may need operator intervention to correct misreading problems.

SUMMARY

(8) At least one embodiment comprises a system for admitting automobiles to a venue comprising: at least one camera for recognizing an automobile, and at least one camera for recognizing a license plate. There can be also at least one server for receiving a picture of the automobile and a picture of the license plate of the automobile. Furthermore, there is at least one database for storing matching and identifying the picture of the automobile and the license plate. In addition, there is at least one screen for allowing a user to identify their automobile taken from a list of automobiles and to make payment for that automobile.
(9) In one embodiment, the camera for recognizing the automobile and the camera for recognizing

the license plate are the same camera.

(10) There is also a process for tracking an automobile in an area. The process can comprise the steps of identifying a predefined area, and next identifying an automobile for screening into the area. Next the process involves photographing the automobile including the body and the color of the automobile. Next, the process involves photographing a license plate on the automobile. Next, the process involves storing the picture of the automobile in a database. Next the process involves storing the picture of the license plate in a database. Next the process involves creating an automobile identity in the database based upon the picture of the license plate and the picture of the automobile. Furthermore, the process involves storing the identity of the automobile in the database. Next the process involves generating an invoice for the automobile and attaching the invoice to the identity of the automobile in the database. Next the process involves selecting a list of automobiles associated with outstanding invoices. Next the process involves presenting a list of identified automobiles having outstanding invoices to the user. Next the process involves having a user select an automobile from a list of automobiles in the database comprising automobiles that have not paid an entry fee.

(11) Thus, there is a powerful, elegant and easy to implement solution to this problem of unidentified vehicles inside the venue. This is achieved by coupling LPC plus machine learning boards with a purpose designed software app, the venue app, which eliminate the troublesome and critical failures of the state-of-the-art LPC-based system alone. In our system, a customer drives through the entrance, without a gate or attendant, at posted speed while LPC reads his vehicle license plate and takes a number of front and rear-view photos of the vehicle.

(12) The patron is not stopped even if the LPC does not register a perfect read, as an image of this entry will be stored. The venue payment system, the payment app, enters each vehicle's information into the UNPAID category list on the venue payment app. The customer parks his vehicle and opens the payment venue app on the venue local network with his portable electronic device and enters his vehicle license plate and makes payment. The customer may also use a public or private network to access the payment app or use one of the payment kiosks situated in the parking area. The Customer may also seek out a Ranger and make payment via the Ranger's mobile app. When the customer completes payment, the payment app moves the customer's data to the PAID category list.

(13) Failure to fully read the license plate information of any entering vehicle results in placing the photos and partial license plate data, in serial format, of such vehicle in the INCOMPLETE category list by the venue payment app. This information will include the serial location of the partial license plate data and thus produce data to narrow the possible identity of each partial read.

(14) If a patron attempts payment and his license plate information has not been fully captured by the LPC, the customer enters his license plate information into the venue payment app and the app displays an array of photos of vehicles from the INCOMPLETE category list which most closely matches the information entered by the customer. The customer can then easily identify his own car to the payment app from this ‘line-up’ and complete making his payment. The payment app then moves his information into the “Paid” category. By this method, essentially all cars which enter this new type of automated entry will make the required payment without the need for an entry gate or toll booth attendants. Using a machine learning board to find best approximate matches between the information entered by customer and the data found in the INCOMPLETE category list allows for a greatly reduced number of possible matches needed on the “line-up” list and makes it very easy for customer to identify his own vehicle on the INCOMPLETE list, even in a very large venue.

(15) Assuming a five percent failure rate by LPC to capture license plate information and a venue with a parking capacity of one hundred cars, the payment app would, at worst, only have to show photos of five vehicles to the customer for identification. The array shows high resolution photos in color with views of the vehicle both from front and back. Identification by the owner of his vehicle

from such a photo array would be quick, easy and absolute. Our system presents near matches of the most likely car to the patron at time of payment. This feature enhances the overall identification efficiency of our system to essentially 100 percent and makes it possible to operate such venues without gates and gate workers. The presentation of these pictures would be sent via an instruction of a server having a microprocessor, accessing a database of saved or archived unpaid vehicles, and then sending this list of vehicles to the user's remote device or kiosk. The array of unpaid vehicles is sorted based upon time, location, date and unpaid status.

(16) Indeed, even in a much larger venue with a parking capacity of a thousand cars, at worst, the photo array would consist of only fifty cars in a five by ten array under the worst possible circumstance. However, failure to identify license plates by LPC is typically a partial failure which results in capturing most of the plate data including sequence and location of identified and missing digits. Therefore, the machine learning board will use the partial identification to match out all license plates which do not fit the partial match. Thus, reading only two digits and their location from a six-digit plate will reduce the “line-up” to but a very few likely prospects, especially if the data includes color photos of the customers car. This system therefore is capable of handling very large venues efficiently.

(17) These venues have Rangers patrolling the parking lots, campgrounds and beaches on motorized vehicles to keep general order and aid the customers in various ways. In the present invention, Rangers will carry a mobile device which has access to all the venue app features and can scan license plates. In this way, a Ranger can scan license plates inside the venue and determine the payment status of each parked vehicle. The ranger can also accept payment from customers through the ranger payment app if the customer so desires. Based on venue rules, the ranger can issue violations to any vehicle which has failed to make payment within the venue-specified time period. Violations are sent to the registered owner of the vehicle by a violation payment service.

(18) The system also uses LPC at points of exit and records vehicles leaving the venue. Collecting this information has several important benefits to the operation of the venue. Keeping a constant tally of incoming and departing vehicles allows the system to determine the current and accurate level of occupancy of the venue. The system can then advise potential visitors by email or other messaging that openings exist. In this way the system can be operated at increased occupancy and higher profitability. The second benefit is that LPC at the exit allows for a second chance at identifying cars which were not captured at entrance. Checking LPC data at exit against the “Unpaid” category list results in a violation being mailed to the owner of record to any violator. The high probability of getting a citation if payment has not been made by a departing vehicle is important in preventing revenue loss or leakage for the venue. A further significant benefit results from combining the power of the venue app with LPC at all points of exit. Venue operators are painfully aware that some unscrupulous customers gain entry by driving into the venue through the exit route and avoid both detection and making payment. Current state of the art systems have no way of dealing with this known financial loss since increasing personnel to watch against this type of violation is not financially viable.

(19) Combining LPC camera at point(s) of exit with the venue app allows for direct detection and identification of such violators. Owners of violating vehicles will be sent notice of violation automatically. Further, a Ranger can be alerted to such an entry violation and dispatched to confront the violator when he parks his vehicle. Ranger can reach the violating vehicle by searching for license plates number on parked cars for the culprit identified by the venue app. Alternatively, a camera located high above the venue can automatically tag and track any vehicle entering through an exit and show the Ranger on his venue app exactly where the perpetrator has parked his vehicle.

(20) As has been made clear, this new entry payment system has many benefits over the state-of-the-art payment systems. The present invention reduces labor costs for the venue by eliminating the

need for entry booth attendants. The system also eliminates congestion at point(s) of entry and the long waiting lines at point of entry. The venue app's ability to keep track of current occupancy levels and its ability to inform potential customers of open venue spots can increase revenue for the venue. The obvious presence of the ‘all-seeing’ LPC based venue app will reduce and likely eliminate revenue loss through nonpayment.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Other technical advantages may become readily apparent to one of ordinary skill in the art after review of the following figures and description.

(2) It should be understood at the outset that, although exemplary embodiments are illustrated in the figures and described below, the principles of the present disclosure may be implemented using any number of techniques, whether currently known or not. The present disclosure should in no way be limited to the exemplary implementations and techniques illustrated in the drawings and described below.

(3) Unless otherwise specifically noted, articles depicted in the drawings are not necessarily drawn to scale.

(4) FIG. **1** is a plan view of a first encounter with the system;

(5) FIG. **2** is a plan view of an encounter with a second system;

(6) FIG. **3** is a block diagram of a layout of a network;

(7) FIG. **4**A is a block diagram of a portable computing component;

(8) FIG. **4**B is a block diagram of a server;

(9) FIG. **5** is a flow chart for the process;

(10) FIG. **6** is a flow chart for the process for tracking the entry and exit of autos from a site;

(11) FIG. **7** is a block diagram of a server layout for connecting to municipal databases;

(12) FIG. **8** is a schematic block diagram for a series of parking lots;

(13) FIG. **9** is a flow chart for registration; and

(14) FIG. **10** is a flow chart for tracking delinquent autos.

DETAILED DESCRIPTION

(15) FIG. **1** is a plan view of a first encounter with the system which shows a predefined area **10** which can be defined by GPS coordinates stored in a database such as database **54***a* (See FIG. **3**). In, addition there is also another pre-area **10**.**1** which is positioned outside of the pre-defined area, and which is configured for pre-screening of an automobile. The GPS coordinates of this area **10**.**1** can also be stored in the database such as database **54***a* in FIG. **3**. There is a station **12**, which is configured to potentially house an attendant, if necessary. An attendant **14** can have an electronic device **15** which can be used to check an individual into the area. Alternatively, a check-in station **48** comprising a camera **27** disposed on a camera stand **25** can be used to take a picture of a license plate **31**. In addition, there is also another check-in station **48**, comprising another camera **28** positioned on another camera stand **26** which is configured to take a picture of a license plate **31**.**1**. Thus, there can be two check in stations, on positioned to take a picture of a front half or front end of an auto and another positioned to take a picture of a back half or back end of an auto. There is also shown an attendant **42** who is positioned behind the automobile with an electronic device which also allows a user to pass into the pre-defined area once the user is screened. The electronic device **36** held by the other attendant **42** can be in the form of a portable camera such as a smartphone wherein the attendant **42** can take a picture of a back license plate **31**.**1** if camera **28** somehow does not recognize or capture license plate **31**.**1**.

(16) There is also a fence **20**, a lift **22** and a gate **24**. The gate **24** is configured to be lifted by the lift **22** upon instructions from either a server or from an attendant after the automobile **34** has been

identified through the license plate picture and/or through the picture of the automobile. Disposed inside of the automobile **34** is a passenger **32** as well as an electronic device **30** which can also be used to identify the individual and/or automobile as well. Disposed inside of the predefined area is a kiosk **17** which includes a base **19** and a screen **16**. The user once inside of the pre-defined area **10** can then work with the kiosk **17** to check in the vehicle if necessary. While two attendants **14** and **42** are shown, with this automated system, it is possible to operate the entrance and entry of an automobile without attendants.

(17) FIG. **2** is a plan view of an encounter with a second system. In this view there are a plurality of stations **48** which are positioned in a line which can be used to identify particular autos such as autos **34** in a line. These stations are configured to take a picture of both a front license plate such as license plate **31** and a back-license plate such as license plate **31.1** as well as a picture of the auto itself. This can be done in an initial region such as region **41**. In this view, there is shown a house or station **12**, an attendant **14**, a predefined area **11**, which has a plurality of parking spots **11***a*, **11***b*, **11***c*, **11***d*, **11***e*, **11***f*, **11***g*, **11***h*, **11***i*, **11***j*, **11***k*, **11***m*, **11***n*, **11***o*. As shown in FIG. **1** this area can have a fence **20** a lift **22**, a gate **24**. A kiosk **17** can be disposed inside of the area while stations **48** for photographing both the auto and the license plate are positioned at both the entrance and at the exit **49** of the area. Wireless emitters such as a WIFI emitter **61**, a cell tower **62** and/or a Bluetooth emitter **63** can be positioned adjacent to these areas to allow for full wireless communication by the operators of the autos or motor vehicles and any attendants or rangers in the area.

(18) FIG. **3** is a block diagram of a layout of a network which can be used to accomplish this system. For example, there is a computing device **18** which can be housed inside of station **12**. This computing device can be connected to a network either in a wired or wireless manner. This computing device **18** is coupled to or in communication with the camera **27** which is part of the station **48**. In addition, computer or computing device **18** is in communication with kiosk **17** comprising base **19** and screen **16**. In addition, there are users' handheld devices **30** and **36** as well as an attendant tablet **15** which are also in communication with the network. The network can also include the internet **56** which is also in communication with a Bluetooth emitter **63**, Wi-Fi emitters **61**, cell towers **62** as well. The Wi-Fi emitter **61**, the cell tower **62**, and/or the Bluetooth emitter **63** are configured to allow for wireless communication between the components such as the wireless devices such as the phones or handhelds **30**, and **36** as well as the first device such as a tablet **15**. Once the remote components such as the mobile devices **15**, **30** and **36** the camera **27** and the kiosk **17** communicate through the internet, this information can be passed through a firewall **55** onto a server network **51** which can include an application server **53**, and a database server **54**, having a database **54***a*. The database is configured to store the information about the site, about each automobile and about each consumer.

(19) While this server system and layout is simply an example, this system can be used as a closed end system so that has a LAN **33** which interfaces with the kiosk, the camera **27** or station **48** the WIFI emitters **61**, the Bluetooth emitters **63** the computing device **18** such that the computing device **18** can run a local application of the program and process without the need for interaction through a network connection such as through the internet **56** onto a firewall **55** or on to the servers **51** including application server **53**, and or database server **54**.

(20) FIG. **4**A is a block diagram of a portable computing component which discloses the standard components associated with any one of the associated components of electronic devices **15**, **17**, **30**, **36** and **48**. For example, this design can include a first phone or pad **15**, a second phone or pad **30**, a third phone or pad **36**, as well as kiosk **17** and a camera **27** of camera station **48**. However, with this design, camera **27** of camera station **48** can also be driven by an optional motor **116** which controls the angle and orientation of this camera. Thus, with this design there is a motherboard **109**, a microprocessor **101**, a memory **102**, a mass storage device such as a hard drive or EEProm, a power supply **104** such as a battery or hard-wired AC power, an optional SIM card or cellular card **105** which allows for communication with cell towers **62**. There is also a GPS **106** which allows for

communication with GPS satellites **60**. In addition, there is a video output **107** as well as a WIFI/TCP/IP transmitter **108**. Furthermore, depending on the device there is an optional video screen **100***a* as well.

(21) FIG. **4**B is a block diagram of a server or computing device such as server application server **53**, database server **54** or computing device **18** which can be associated with the station **12**. With this design, there is a motherboard **209**, which houses a microprocessor **201**, a memory **202**, a mass storage **203**, a power supply **204** for powering the components, a video output **205**. (optional). There can also be an I/O port (input/output port) and a communications port comprising a TCP/IP/WIFI port **208** allowing the device to communicate with the network.

(22) FIG. **5** is a flow chart for the process for tracking an automobile in an area. The process is mostly if not entirely performed using microprocessor **201** in coordination with memory **202**, and mass storage **203** to add or modify information in database **54***a* in database server **54**, as it is controlled by application server **53**. The process can comprise the steps of step **1** identifying a predefined area. The identification of the pre-defined area can include a remote device such as a tablet **15**, or a portable phone **30** or **36** having a GPS transmitter. Devices can be used to create a boundary for the area using GPS coordinates that can be stored in the database such as database server **54**. In addition, each of the stations such as station **48** or the kiosk can also be located on a GPS based plot map of the pre-defined area. In that way each station both at the entrance and the exit can be identified based upon either its location or based upon a MAC address or other network identifying feature. Next, in step **2** an attendant or the system can identify an automobile for screening into the area. Next in step **3***a* the process involves photographing a license plate (This occurs using a camera **27** of the station **48**. The camera can be moved such as through motor **116** to obtain the proper picture of the license plate and the body of the automobile (Step **3***b*). The body is identified via the camera by recognizing the shape and relative size of the body. Next, in step **4** the process involves storing the picture of the license plate and or read text of the license plate in a database. In this case, the picture of the license plate is read and transcribed into digits or typographic indicia based upon the analysis of the shape and arrangement of the pixels on the picture of the license plate which is then stored in the database.

(23) Thus, the picture from station **48** is transmitted through an associated transceiver **108** through the internet **56** through firewall **55** into application server **53** and then stored in database **54***a* of database server **54**. Alternatively, station **48** transcribes the picture of the license plate into a series of digits (numbers and/or letters) or other indicia which is then transmitted and then stored in the database. Alternatively, the station **48** can read the indicia on the license plate picture, transcribe the information on the license plate and then forward the transcribed information to the server. For example, if the picture revealed that the plate was a New York State Plate based upon coloring, and logos, it would then read the indicia on the plate such as ABC-1234, which might be a standard plate set of indica conforming to a series of letters followed by a series of numbers. The plate may also have indicia or lettering indicating the state that the plate originates from or a logo such as the statue of liberty or other type of logo to indicate the state in which the plate is from. The reader can also read the coloring of the plate as well.

(24) Next in step **5**, the process involves storing the picture of the auto in a database such as database **54***a*. While all of the steps are optional this step is optional and not in the preferred embodiment. Next, in step **6**, the process involves creating an automobile identity in the database based upon the picture of the license plate and the picture of the automobile. The creation of the identity can involve using microprocessor **201** running a program in memory **202** fed from mass storage **203** to assemble the photographs of the license plate, the indicia of the license plate (digits etc.), the automobile and any known profiles of the auto into the database. Alternatively, it could simply be the data extracted from a picture of the license plate or of data entry of the license plate itself if necessary. If the license plate of an automobile is identified as a past user, then this picture or data is matched with a previous profile that is already stored in the system. If the license plate is

new, then the user can fill out this profile and pay for the entrance in a later step via the kiosk **17**. In addition to setting up an identity of the automobile based upon the license plate information, the process can also include the setting up or registering of a user (See FIG. **9** for an example of the process for registering a user).

(25) Thus, once the profile is created or at least initially set up, in step **7** the process involves storing the identity of the automobile in the database.

(26) Next, in step **8** the system would maintain a list of unpaid invoices for presentation and accounting for future payments. This list of unpaid invoices would be stored in the database and then presented as available for review on the kiosk.

(27) Next, in step **9** the patron presents their license plate and payment to the kiosk operator or to the kiosk alone, or on their mobile device such as via a webapp. This presentation of the license plate then matches the user or patron with the list of unpaid invoices so that the user can then pay their invoice.

(28) As the patron is inputting their license plate for future payment, the patron can, in step **10** the patron may choose to register with the system for further simplification of the process at next use. For example, this process is shown in greater detail in FIG. **9**.

(29) Next, in step **11** the system assesses payments, updates the list of associates payments to vehicles in the unpaid list and removes vehicles from the unpaid list when a payment is made. Thus, with each punch in of information, the field of available autos can be narrowed in the database until only the remaining matching auto with the matching license plate is available. If the numbers do not exactly match, then the process proceeds to step **12**.

(30) Next, in step **12**, if the system does not find the license plate information input by the patron, the system uses fuzzy logic (artificial intelligence) to pair the plate that was entered with partial or full reads of the license plates. This fuzzy logic includes matching a partial list of indicia or digits stored in the database regarding the outstanding invoices with the list of digits or indicia input by the user. With each input by the patron, the list of available license plates from the list of unpaid invoices is narrowed down until the only remaining plates are left for payment.

(31) Next, in step **13** the user or patron selects one of the vehicles from the list and makes a payment. The vehicle is then removed from the list. Next, as indicated above, with payment, the system matches the payment with the plate listed or uses fuzzy logic to pair the plate that was paid with the list of plates in the unpaid list. Next, in step **15** unpaid vehicles are selected and presented to the supervisor for violations processing.

(32) Next in step **16**, the system such as through microprocessor **201** can log when a user leaves the pre-defined area. This allows for the system to open up new spots for autos to enter into the predefined area. The confirmation that the user leaves the area can take place when a camera from an associated station such as station **48** at an exit takes a picture of the leaving patron's license plate and then this license plate is matched with one of the license plates of the recent entrants into the predefined area.

(33) FIG. **6** is a flow chart for the process for tracking the entry and exit of autos from a site. For example, the process starts with step **21** wherein the system tracks the entry of an auto via a license plate or other identifier. In this step the system conducts an optical character recognition scan for the characters in the license plate. Each of the individual characters is recognized and indexed to match based upon shape with characters and shape kept on file. This step is performed by the microprocessor such as microprocessor **101** or **201** recognizing each individual shape of characters in a license plate. Next, upon recognition of the license plate and the auto, the auto is allowed into an area. In this step, even if the auto is not fully recognized it can still be allowed into an area. Next, in step **23**, a database is scanned to determine whether the auto is registered. This database can be taken from any suitable database such as from a first DMV server **130**, or an alternative DMV server **131** (See FIG. **7**). Alternatively, a central server **125** can pull this information from commercial databases such as commercial database **132** or commercial database **133**. Central

server **125** then feeds this information into server networks **51***a* or **51***b* which are similar to the server network **51** shown in FIG. **3**. In step **24** the system determines that if there is no match the system flags the entry to determine and indicate that the auto is not identified. Next, in step **25** the system such as central server **125** notifies the system controller (administrator) that the auto is not registered and then the system provides for a location of the auto based upon the scanning or check in station **48**. Each check in station has an identity and an address so that both the central server **125** as well as the server network shown in FIG. **3** would have a record of the location of this check in station **48**. The address can be stored as GPS coordinates or as an actual address in a database such as database **54**A. Next, in step **26**, the system can then send out a message and notify a field agent to find a non-identified auto.

(34) Either before or after step **26**, step **26***a* can occur wherein a user can be registered. For a more complete description of registering a user see FIG. **9** for the process for registering a user.

(35) Next, in step **27** the system can scan the auto upon leaving the site, match the leaving auto with the entering auto and then determine whether this was one of the missing non-identified autos.

(36) FIG. **7** is a schematic block diagram for a distributed network of servers in a system. For example, there is a central server **125** which is coupled to server networks **134** and **135** which are representative of servers **51***a* and **51***b* (see server network **51** in FIG. **3**. Central server **125** is also in communication with different commercial databases **132** and **133** which may have information from commercially available sites to track patrons or users to make sure that unpaid bills are processed and paid as well as unidentified users/patrons are tracked down for a later processing of payment. In addition, the central server **125** is also in communication with DMV server **130** and or DMV server **131**. Thus, central server **125** can then communicate with these DMV servers as well to obtain the identity of the license plate holders so that the central server **125** can be used to then identify and invoice personally the owners of the vehicle based upon either commercial information stored in a commercial database **132** and **133** or from DMV servers **130** and **131**.

(37) FIG. **8** is a schematic block diagram for a series of parking lots. With the entry and exit system comprising at least the camera station **17** as well as the check in station **48** each of the lots **151**, **152**, **153**, **154**, **155**, and **156** can then track each of the cars. When the first lot **151** fills up, the system would know this from tracking the autos in an entry and exit manner. Then the system could notify an attendant that the cars/autos should be directed towards the next lot. When the second lot fills up the next set of cars/autos are sent onto the next lot and so on. In this way the lots can be filled sequentially and the system such as central server **125** can track the occupancy of the vehicles as well.

(38) One way to track the autos/users is to have an extensive registration system for each of the autos/users. The user can either pre-register or register on site through check in station **48**. For example, FIG. **9** is a flow chart for registration of a user/auto, wherein in this embodiment, the user is invited to register in step **901**. This registration includes inserting the name and address in step **902**. Next, the user can insert his/her telephone number in step **903**. Next, in step **904** the user can insert his/her email address in step **904**. Next in step **905** the user can insert his/her make and model of the car. Next, in step **906** the user can insert his or her color of the car. Next, in step **907** the user can insert his/her license plate of the car. Next, in step **908** the user can insert a picture of the car. Next in step **909** the user can insert a picture of the user as well for the user's profile. With this composite profile of the user, if a license plate is not properly read, or identified upon entry or exit, the system can match a partial reading of the license plate with the existing database of registered users. For example, if the auto had a series of identifying dents or marks, the recent entry photograph can be matched with the photograph on file of the registration. Furthermore, the color, and/or the make/model of the auto, can then be matched with the information on file so that if there is only a partial reading of the information, this partial reading can be matched with the other identifying characteristics and then be used to identify the auto.

(39) FIG. **10** is a flow chart for tracking delinquent autos, which shows that in step **1001** a portion

of the license plate can be matched with the database to generate a partial match. Next, in step **1002**, the system can match the color of the auto with the database and match the color with the partial list of the license plate. Next, in step **1003** the system identifies dents on the auto via pictures of the auto. This step can be based upon identifying the unique shapes in the body wherein the dents are determined to deviate in shape from a surface on the body.

(40) Next in step **1004** the system can identify the car owner or driver via facial recognition. Next in step **1005** the system can identify other indicia or characteristics such as the make/model of the auto. These identifying features are matched only with license plates that have at least partial readings of the same indicia such as the same letters and numbers in sequence. Thus, the system narrows the query for color, make/model or dents to only those autos that are in the pre-selected list of qualifying partial license plate matches.

(41) In at least one embodiment, the step of presenting a list of identified automobiles associated with outstanding invoices (step **10**) comprises presenting the list on a kiosk inside of the pre-defined area.

(42) In at least one embodiment, the step of tracking when an automobile leaves the pre-defined area (step **12**) comprises taking a picture of the automobile body and taking a picture of the automobile license plate and matching these pictures with pictures stored in the database.

(43) Modifications, additions, or omissions may be made to the systems, apparatuses, and methods described herein without departing from the scope of the disclosure. For example, the components of the systems and apparatuses may be integrated or separated. Moreover, the operations of the systems and apparatuses disclosed herein may be performed by more, fewer, or other components and the methods described may include more, fewer, or other steps. Additionally, steps may be performed in any suitable order. As used in this document, “each” refers to each member of a set or each member of a subset of a set.

(44) To aid the Patent Office and any readers of any patent issued on this application in interpreting the claims appended hereto, applicants wish to note that they do not intend any of the appended claims or claim elements to invoke 35 U.S.C. 112(f) unless the words “means for” or “step for” are explicitly used in the particular claim.

(45) Although specific advantages have been enumerated above, various embodiments may include some, none, or all of the enumerated advantages.

Claims

1. A process for tracking an automobile in an area comprising the steps of: identifying a predefined area; identifying an automobile for screening into the area; photographing the automobile to obtain a picture of the automobile including a body and a color of the automobile; photographing a license plate on the automobile to obtain a picture of the license plate; storing the picture of the automobile in a database; storing the picture of the license plate in the database; creating an automobile identity in the database based upon the picture of the license plate and the picture of the automobile; storing the identity of the automobile in the database; generating an invoice for the automobile and attaching the invoice to the identity of the automobile in the database; selecting a list of automobiles associated with outstanding invoices; receiving license plate information from a user; retrieving a profile of the user based on at least a partial match of the received license plate information; retrieving an image of a user automobile associated with the user from the profile; retrieving an image of an owner of the user automobile from the profile; taking a picture of the user and determining that the user matches the owner using facial recognition; identifying, from among automobiles on the list of automobiles associated with outstanding invoices, candidate automobiles having license plate information that partially matches the received license plate information; matching identifying features of the candidate automobiles with identifying features of the user automobile based on the image of the user automobile; presenting a list of identified automobiles

having the matched identifying features to the user, wherein the matched identifying features comprise a make and model of the user automobile, a color of the user automobile, and dents on the user automobile; and having the user select an automobile from the list of the identified automobiles for payment of an entry fee.

2. The process as in claim 1, wherein said step of identifying a pre-defined area comprises outlining GPS coordinates for said area and storing said GPS coordinates in the database.

3. The process as in claim 1, wherein said step of presenting a list of identified automobiles associated with outstanding invoices comprises presenting the list on a kiosk inside of the predefined area.

4. The process as in claim 3, further comprising the step of tracking when the automobile leaves the predefined area.

5. The process as in claim 4, wherein said step of tracking when the automobile leaves the pre-defined area comprises the picture of the automobile and the picture of the license plate with pictures stored in the database.

6. The process as in claim 4, further comprising the steps of: contacting an external database for the license plate information; and identifying an owner of the automobile based upon the license plate information; contacting the owner to notify the owner that the automobile has left a site without paying for a service; and forwarding an invoice to the automobile owner so that the owner pays the invoice.

7. The process as in claim 6, wherein the step of forwarding an invoice to the automobile owner comprises sending the invoice via electronic transmission.

8. The process as in claim 7, wherein the step of forwarding an invoice to the automobile owner comprises sending the invoice via email.

9. The process as in claim 7, wherein the step of forwarding an invoice to the automobile owner comprises sending the invoice via text messaging.

10. The process as in claim 6, wherein the step of identifying an owner of the automobile based upon the license plate information comprises scanning an image of the license plate, and then determining the characters on the license plate via text recognition software and then comparing the characters on the license plate to a set of characters stored in an external database.

11. The process as in claim 1, wherein said step of matching identifying features comprises matching a make and model of an auto with at least a partial reading of a license plate.

12. The process as in claim 11, wherein the step of matching identifying features comprises matching dents on an auto with stored pictures of the auto.

13. The process as in claim 12, further comprising the step of tracking the location of the automobile.

14. The process as in claim 13, further comprising the step of determining the location of the auto via determining a location of a scanning check in station which checked in the automobile.

15. The process as in claim 13, further comprising the step of selectively sending a field agent to find the location of the automobile.

16. The process as in claim 1, further comprising the step of determining how many automobiles are in a lot by determining how many automobiles have entered and left the lot.

17. The process as in claim 16, further comprising the step of directing additional automobiles to an adjacent lot when a first lot is filled up.